Table 1: Drivers

DriverID	DriverName	VehicleType	ContactNumber	JoinDate	BaseSalary
1	Ahmed Ali	Truck	01711223344	2019-05-12	28000
2	Bablu Mia	Van	01822334455	2021-08-20	18000
3	Charles Rozario	Heavy Truck	01733445566	2018-11-05	35000
4	David Hasan	Van	01944556677	2022-02-15	16500
5	Emon Tariq	Mini-Truck	01555667788	2023-01-10	15000
6	Farhan Khan	Truck	01766778899	2020-07-22	26000
7	Golam Rabbi	Heavy Truck	01877889900	2019-09-30	33000
8	Hasan Mahmud	Mini-Truck	01788990011	2021-12-12	17000
9	Imran Hossain	Van	01999001122	2022-05-18	16000
10	Jamal Uddin	Truck	01700112233	2020-03-25	27500

Table 2: Vehicles

VehicleID	DriverID	LicensePlate	VehicleModel	PurchaseYear
V01	1	DHK-1122	Tata Ultra	2018
V02	2	SYL-3344	Mahindra	2020
V03	3	CTG-5566	Ashok Leyland	2017
V04	4	RAJ-7788	Suzuki Super	2021
V05	5	KHL-9900	Tata Ace	2022
V06	6	DHK-2233	Eicher Pro	2019
V07	7	CTG-4455	BharatBenz	2018
V08	8	BAR-6677	Mahindra Maxx	2020
V09	9	SYL-8899	Suzuki Super	2021
V10	10	RAJ-0011	Tata Ultra	2019

Table: Triplogs

TripID	DriverID	DestinationCity	DistanceKM	TripFare	TripDate
101	1	Dhaka	250	5000	2023-10-01
102	2	Sylhet	320	6000	2023-10-02
103	1	Chattogram	450	8500	2023-10-05
104	6	Rajshahi	280	5500	2023-10-06
105	3	Dhaka	250	5200	2023-10-08
106	7	Khulna	380	7000	2023-10-09
107	2	Sylhet	320	6000	2023-10-12
108	8	Barishal	210	4500	2023-10-14
109	10	Rajshahi	280	5400	2023-10-15
110	6	Chattogram	450	8300	2023-10-18

Section A

Varendra University

Department of Computer Science and Engineering

5th Semester Mid Lab Test (Spring-2026)

Course Code: CSE 3104

Course Title: Database Management System Lab

Marks: 30

e: 45 minutes

uestion o	CO's no.	Description	Taxonomy domain/level	POs	K	P	A
1	CO1	*Utilize SQL commands to perform basic database tasks*	*Cognitive/ Apply*	*PO1*	*WK4*	-	-

Marks

1. **Modify** the Drivers table using an SQL query to increase the BaseSalary by 12% for all drivers who operate a 'Van' and joined the company before the year 2022. 6
2. **Calculate** the total months of experience for each driver using their JoinDate and the current system date. Display this calculated value alongside their DriverName converted entirely to uppercase letters. 6
3. **Implement** a query using mathematical functions to calculate a 15% festival bonus based on the TripFare for all trips that exceeded 300 KM. Round the calculated bonus to the nearest whole number and display it along with the TripID and DestinationCity. 6
4. **Demonstrate** the use of aggregate functions to find the total number of trips made and the maximum TripFare charged for each DestinationCity. Apply a filter to show only those cities where the maximum trip fare is strictly greater than 5500. 6
5. **Execute** the necessary SQL commands to alter the existing Drivers table to enforce data integrity. You need to add a unique constraint to the ContactNumber column and a check constraint to ensure the BaseSalary is never lower than 10,000. 6